# Tofa Kimani Mwangi

**Software Engineer — Full-Stack, AI, Data & IT**

Full-Stack & Backend | AI & LLM Engineering | Data Engineering & Analytics | IT Support | DevOps

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## PROFESSIONAL SUMMARY

Software engineer and Applied Computing graduate working across full-stack development, backend systems, AI and LLM integration, data engineering, analytics, IT support and deployment. Built and deployed Agri-Link, an ERP and point-of-sale system now in trial at a Kenyan agrovet (Java, Spring Boot, PostgreSQL, Next.js); built Job Agent, an AI job-search assistant integrating an LLM (Claude) with validated structured outputs; and engineered production-style ETL pipelines with PySpark, Apache Airflow and Docker. ALX-certified Data Engineer, ready to step into software engineering, AI engineering, full-stack, backend, data, IT support or DevOps roles.

## TECHNICAL SKILLS

- **Programming Languages**: Python, Java, JavaScript, TypeScript, SQL, Bash, C#.
- **Backend & API Engineering**: Spring Boot, Node.js, Express, FastAPI, Pydantic, RESTful APIs, JWT Authentication, OAuth 2.0, Role-Based Access Control (RBAC), Flyway Migrations.
- **AI & LLM Engineering**: LLM Integration (Claude API), Prompt Engineering, Structured Outputs (JSON Schema + Pydantic), AI Output Validation, Prompt-Injection Defence, Prompt Caching.
- **Frontend**: Next.js, React, TypeScript, Tailwind CSS, HTML/CSS, Bootstrap.
- **Big Data & Data Engineering**: PySpark, Apache Spark, Apache Airflow, ETL/ELT Pipelines, Data Warehousing, Dimensional Modeling (Star Schema), BigQuery, Pandas, NumPy.
- **Data Analytics**: Power BI, Advanced Excel, Matplotlib, Jupyter Notebook, Data Cleaning & Validation, Reporting.
- **Databases**: PostgreSQL (incl. Row-Level Security), MySQL, MongoDB, SQLAlchemy, Database Indexing, Query Optimization, Schema Normalization.
- **DevOps, Testing & Deployment**: Docker, Docker Compose, Linux, Git/GitHub, CI/CD (GitHub Actions), Vercel, Testcontainers, Playwright.
- **Architecture & Data Quality**: System Design, RESTful APIs, Monolith Architecture, Multi-Tenant Data Isolation, Concurrency Control, Idempotency, Double-Entry Ledger Integrity, Pipeline Testing & Quality Gates, Schema Enforcement, Lineage & Auditability.
- **IT Support**: IT Support, Networking, Hardware/Software/Network Troubleshooting.
- **Soft Skills**: Critical Thinking, Cross-Functional Collaboration, Problem-Solving & Troubleshooting, Technical Communication, Requirements Gathering, Attention to Detail, Time Management, Stakeholder Alignment, Adaptability, Agile Methodology.

## WORK EXPERIENCE

**Full-Stack Software Engineer — Agri-Link ERP & Point of Sale** *(Freelance)*

*Agrovet shop, Kenya* *2026 – Present*

- Gathered requirements with the owner and staff and designed an ERP and point of sale covering the till, stock and warehouses, purchasing, customer accounts and a weekly day-old-chick booking cycle; now in trial in the shop.
- Built the backend in Java 21 and Spring Boot with PostgreSQL 16: 35 Flyway migrations, JWT authentication, role checks on every route and row-level security isolating each business's data.
- Built the frontend in Next.js, React and TypeScript around one spreadsheet-style grid used by every screen, so staff work the way they did in Excel (keyboard navigation, paste from Excel, filters, Excel/CSV export).
- Posted every sale, payment and delivery as a balanced double-entry transaction to an append-only ledger; used idempotency keys and row locks so money and stock stay correct under real use; covered it with 106 backend tests on real PostgreSQL (Testcontainers) and end-to-end Playwright tests, deployed with Docker Compose.

github.com/TofaKM/Agri-Link-Showcase

**Corporate Governance Data Analyst – Virtual Assistant**

*Minerva Analytics (Remote)* *January 2026 – April 2026*

- Audited public company filings, annual reports, and proxy statements to log board structure changes, executive leadership shifts, and committee appointments.
- Captured and verified financial metrics, director independence data, and board governance updates with high accuracy across global markets.

**IT Intern**

*Social Health Authority, Nanyuki, Kenya* *September 2024 – December 2024*

- Cleaned and prepared large-scale datasets using Excel and internal diagnostic tools, reducing reporting errors and optimizing data integrity.
- Conducted operational data analysis to identify bottlenecks and collaborated with cross-functional teams to redesign IT workflows, accelerating report generation.
- Resolved technical hardware, software, and network issues across computer systems to ensure high uptime for critical user applications.

## PROJECTS

**Job Agent — AI Job-Search Assistant**
*Python, FastAPI, PostgreSQL, SQLAlchemy, Next.js, TypeScript, Claude API, Docker Compose* *October 2026*

- Built a local AI job-search agent that collects jobs daily from five sources, including LinkedIn job-alert emails read through the Gmail API with OAuth 2.0, then normalizes, deduplicates (MinHash/LSH) and screens them.
- Integrated an LLM (Claude) using JSON-schema structured outputs validated with Pydantic, a retry with error feedback, prompt caching and a daily call budget; job postings are handled as untrusted input to resist prompt injection.
- Kept AI output grounded: code computes the match score from the model's judgements, rejects any cited CV line that doesn't exist, and checks generated CVs and cover letters for numbers or skills not in the source before rendering PDFs.
- Covered the system with 381 automated tests (pytest, saved fixtures, mocked HTTP) and a golden-set evaluation that compares AI recommendations with my own verdicts.

*Private repository — walkthrough on request*

**Zuri Mobile Customer Churn ETL Pipeline**
*PySpark, Apache Airflow, PostgreSQL, Docker Compose* *June 2026*

- Engineered a production-style telecom churn pipeline processing 10,000 records using PySpark for data validation, deduplication, and standardization.
- Designed a PostgreSQL star-schema data warehouse to optimize dimensional modeling and analytics performance.
- Orchestrated automated, repeatable staging-to-live data workflows using Apache Airflow and Docker Compose.

github.com/TofaKM/zuri-mobile-churn-pipeline

**Hospital Data ETL Pipeline**
*Docker, Python, SQL, ETL Architecture* *June 2026*

- Developed a containerized ETL pipeline in Docker to automate healthcare data ingestion, cleaning, and transformation.
- Structured end-to-end data workflows generating high-precision datasets for operational analysis and clinical reporting.
- Automated routine transformations to allow non-technical stakeholders direct access to accurate business insights.

github.com/TofaKM/etl_hospital_pipeline

**Chap Mall E-Commerce Platform**
*Next.js, Node.js, PostgreSQL, RESTful APIs* *July 2025*

- Architected a multi-tier e-commerce platform featuring Role-Based Access Control (RBAC) across customer, vendor, and admin roles.
- Engineered RESTful APIs and optimized PostgreSQL database queries to handle complex relational data with high data integrity.
- Built inventory tracking logic for live availability updates and implemented secure authentication to lower page latency.

github.com/TofaKM/Chap-Mall-E-commerce-Website-

## CERTIFICATIONS

- **Data Engineer Certificate** – ALX Africa (Aug 2026) — ETL/ELT Pipelines, Data Warehousing, Airflow, Spark, Dimensional Modeling, SQL/Python Automation.
- **Python for Data Science** – ALX Africa (Apr 2026) — Pandas, NumPy, SciPy, Automated Data Extraction, Statistical Hypothesis Testing.
- **Data Analytics Certificate** – ALX Africa (Sep 2025) — MySQL Querying, Power BI Dashboarding, Data Visualization, Professional Reporting, Excel.
- **Introduction to Machine Learning** – Cisco & IBM (Feb 2025) — Supervised/Unsupervised Learning, Model Evaluation, Predictive Analytics (Python).
- **Data Analytics** – Cisco Academy (Aug 2024) — Data Lifecycle Management, Data Cleaning, Advanced Excel, SQL Analysis.

## EDUCATION

**B.Sc. Applied Computing (Computer Science)**
*KCA University, Nairobi, Kenya* *September 2020 – November 2025*

- Core Coursework: Data Structures & Algorithms (DSA), Systems Analysis & Design, Database Management Systems (DBMS), Relational Database Normalization, SDLC.
- Capstone Project: Engineered a full-stack e-commerce application using JavaScript, Node.js, React, and MySQL featuring optimized relational schemas and Role-Based Access Control (RBAC).

**High School Certificate**
*Ndururumo High School, Nyahururu, Kenya* *January 2016 – November 2019*

- Achieved Mean Grade B+ (71 points), focusing on Mathematics, Physics, and Computer Studies.
- Served as Academic Secretary, leading peer tutoring initiatives and managing academic schedules.

## INTERESTS

**Professional Interests & Research**: System Architecture, AI & ML, Tech News, Cloud Data Architecture, Real-Time Data Streaming, Distributed Systems & Microservices
**Activities & Hobbies**: Football, Badminton, Outdoor Activities, Gardening, Animations

## REFERENCES

- **Francis Ngugi** — *Software Engineer* | +254 707 574 430 | francisngugi518@gmail.com
- **Dacosta Otekoh** — *Human Resources (HR)* | +254 701 563 588 | dacostaotekoh@gmail.com